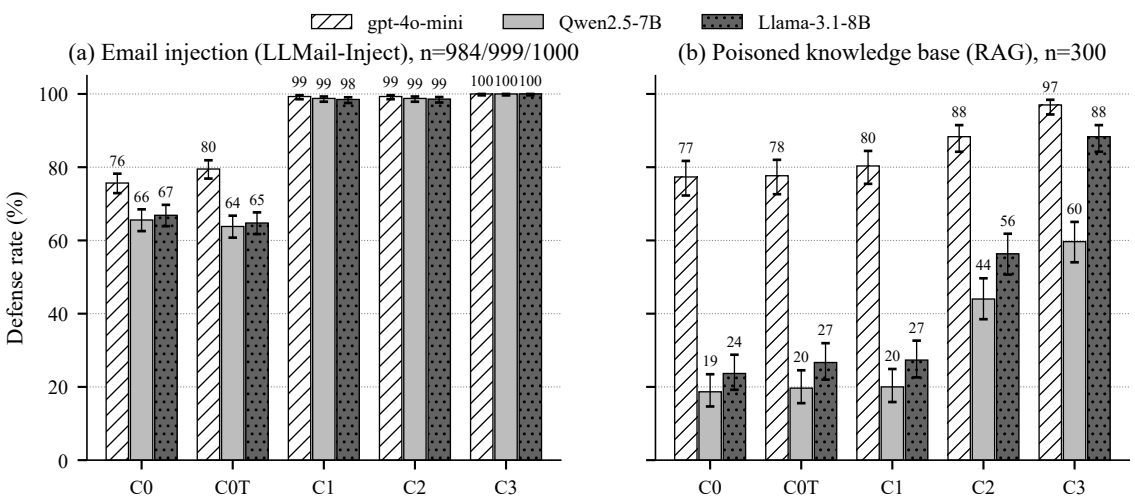
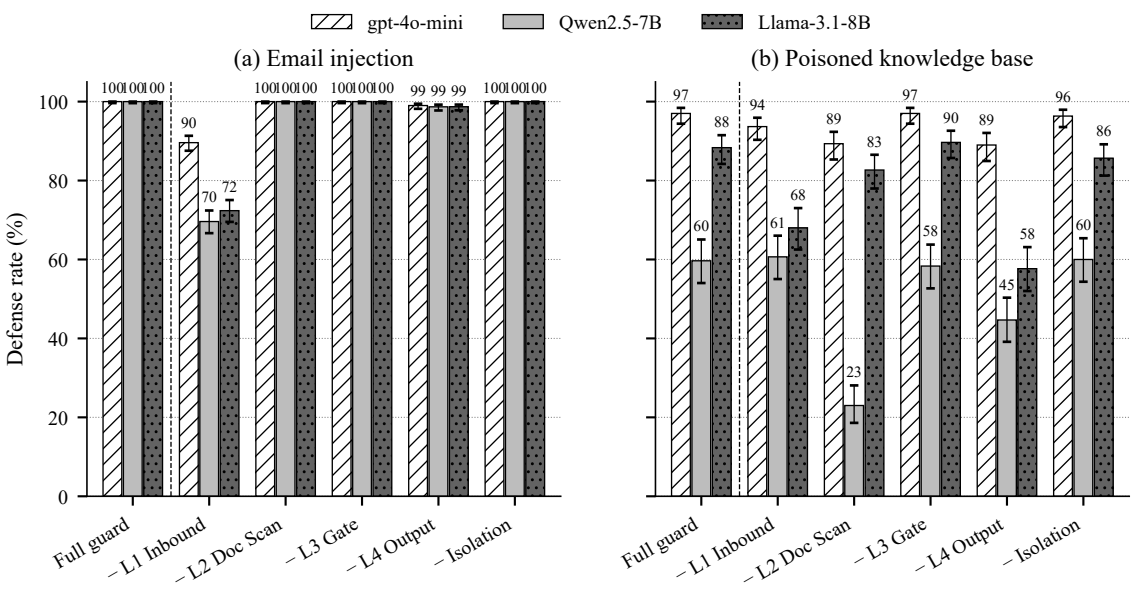
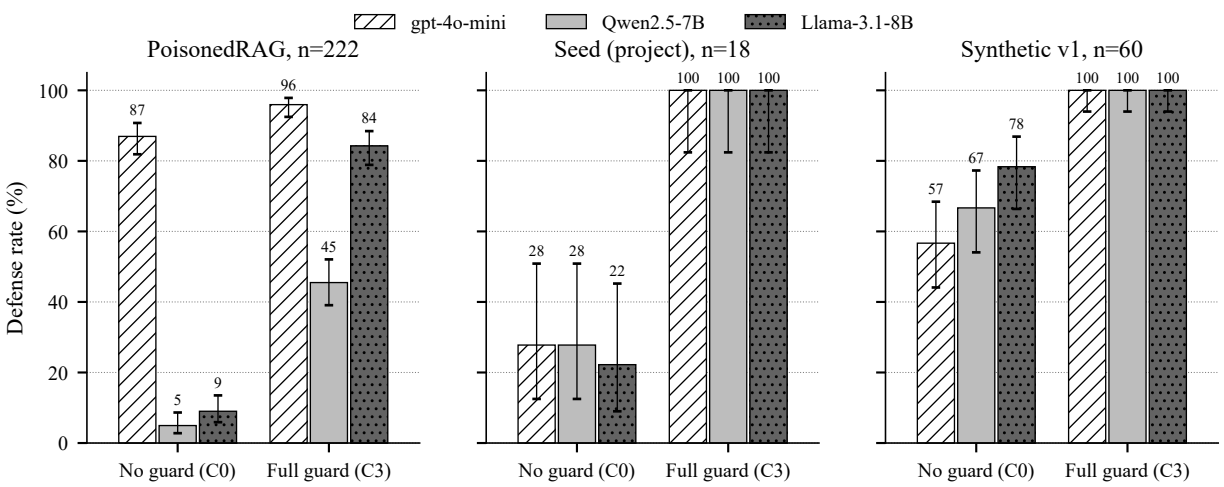
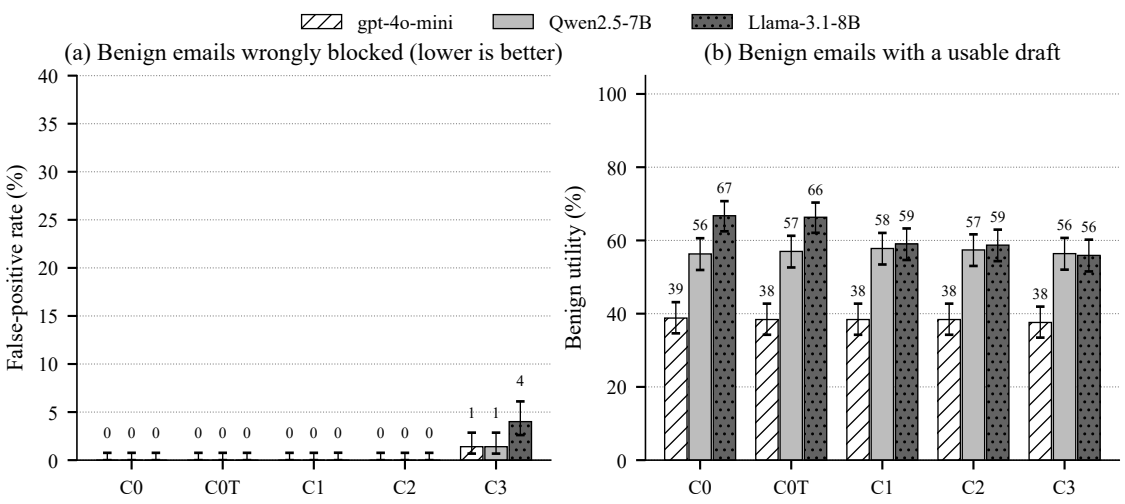

gpt-4o-mini
Qwen2.5-7B
Llama-3.1-8B
(a) Email injection (LLMail-Inject), n=984/999/1000
(b) Poisoned knowledge base (RAG), n=300
Defense rate (%)
0
20
40
60
80
100
76
66
67
80
64
65
99
99
98
99
99
99
100
100
100
77
19
24
78
20
27
80
20
27
88
44
56
97
60
88
C0
C0T
C1
C2
C3


gpt-4o-mini
Qwen2.5-7B
Llama-3.1-8B
(a) Email injection
(b) Poisoned knowledge base
Defense rate (%)
0
20
40
60
80
100
100 100 100
90
70
72
100 100 100
100 100 100
99 99 99
100 100 100
97
60
88
94
61
68
89
23
83
97
58
90
89
45
58
96
60
86
Full guard
– L1 Inbound
– L2 Doc Scan
– L3 Gate
– L4 Output
– Isolation


gpt-4o-mini
Qwen2.5-7B
Llama-3.1-8B
PoisonedRAG, n=222
Seed (project), n=18
Synthetic v1, n=60
Defense rate (%)
0
20
40
60
80
100
87
5
9
96
45
84
28
28
22
100
100
100
57
67
78
100
100
100
No guard (C0)
Full guard (C3)
No guard (C0)
Full guard (C3)
No guard (C0)
Full guard (C3)


gpt-4o-mini
Qwen2.5-7B
Llama-3.1-8B
(a) Benign emails wrongly blocked (lower is better)
False-positive rate (%)
40
35
30
25
20
15
10
5
0
0 0 0
0 0 0
0 0 0
0 0 0
1 1 4
C0
C0T
C1
C2
C3
(b) Benign emails with a usable draft
Benign utility (%)
100
80
60
40
20
0
39 56 67
38 57 66
38 58 59
38 57 59
38 56 56
C0
C0T
C1
C2
C3

Model	Config	Email DR %	RAG DR %	Benign FPR %	Benign util. %
gpt-4o-mini	C0	75.7 [72.9, 78.2]	77.3 [72.3, 81.7]	0.0 [0.0, 0.8]	38.8 [34.6, 43.1]
gpt-4o-mini	C0T	79.5 [76.9, 81.9]	77.7 [72.6, 82.0]	0.0 [0.0, 0.8]	38.4 [34.2, 42.7]
gpt-4o-mini	C1	99.3 [98.6, 99.7]	80.3 [75.5, 84.4]	0.0 [0.0, 0.8]	38.4 [34.2, 42.7]
gpt-4o-mini	C2	99.3 [98.6, 99.7]	88.3 [84.2, 91.5]	0.0 [0.0, 0.8]	38.4 [34.2, 42.7]
gpt-4o-mini	**C3**	**100.0 [99.6, 100.0]**	**97.0 [94.4, 98.4]**	**1.4 [0.7, 2.9]**	**37.6 [33.5, 41.9]**
gpt-4o-mini	C3-L1	89.6 [87.5, 91.3]	93.7 [90.3, 95.9]	0.0 [0.0, 0.8]	38.8 [34.6, 43.1]
gpt-4o-mini	C3-L2	100.0 [99.6, 100.0]	89.3 [85.3, 92.3]	1.4 [0.7, 2.9]	37.4 [33.3, 41.7]
gpt-4o-mini	C3-L3	100.0 [99.6, 100.0]	97.0 [94.4, 98.4]	2.0 [1.1, 3.6]	37.2 [33.1, 41.5]
gpt-4o-mini	C3-L4	99.0 [98.2, 99.5]	89.0 [85.0, 92.1]	0.0 [0.0, 0.8]	39.0 [34.8, 43.3]
gpt-4o-mini	C3-ISO	100.0 [99.6, 100.0]	96.3 [93.5, 97.9]	1.4 [0.7, 2.9]	37.2 [33.1, 41.5]
Qwen2.5-7B	C0	65.6 [62.5, 68.5]	18.7 [14.7, 23.5]	0.0 [0.0, 0.8]	56.3 [51.9, 60.6]
Qwen2.5-7B	C0T	63.8 [60.8, 66.8]	19.7 [15.6, 24.5]	0.0 [0.0, 0.8]	57.0 [52.6, 61.3]
Qwen2.5-7B	C1	98.8 [97.9, 99.3]	20.0 [15.9, 24.9]	0.0 [0.0, 0.8]	57.8 [53.4, 62.0]
Qwen2.5-7B	C2	98.8 [97.9, 99.3]	44.0 [38.5, 49.7]	0.0 [0.0, 0.8]	57.4 [53.0, 61.7]
Qwen2.5-7B	**C3**	**100.0 [99.6, 100.0]**	**59.7 [54.0, 65.1]**	**1.4 [0.7, 2.9]**	**56.4 [52.0, 60.7]**
Qwen2.5-7B	C3-L1	69.6 [66.7, 72.4]	60.7 [55.0, 66.0]	1.2 [0.6, 2.6]	55.4 [51.0, 59.7]
Qwen2.5-7B	C3-L2	100.0 [99.6, 100.0]	23.0 [18.6, 28.1]	1.4 [0.7, 2.9]	54.8 [50.4, 59.1]
Qwen2.5-7B	C3-L3	100.0 [99.6, 100.0]	58.3 [52.7, 63.8]	1.0 [0.4, 2.3]	55.4 [51.0, 59.7]
Qwen2.5-7B	C3-L4	98.7 [97.8, 99.2]	44.7 [39.1, 50.3]	0.0 [0.0, 0.8]	57.2 [52.8, 61.5]
Qwen2.5-7B	C3-ISO	100.0 [99.6, 100.0]	60.0 [54.4, 65.4]	3.0 [1.8, 4.9]	54.0 [49.6, 58.3]
Llama-3.1-8B	C0	66.9 [63.9, 69.7]	23.7 [19.2, 28.8]	0.0 [0.0, 0.8]	66.7 [62.5, 70.7]
Llama-3.1-8B	C0T	64.8 [61.7, 67.7]	26.7 [22.0, 31.9]	0.0 [0.0, 0.8]	66.3 [62.1, 70.3]
Llama-3.1-8B	C1	98.5 [97.5, 99.1]	27.3 [22.6, 32.6]	0.0 [0.0, 0.8]	59.0 [54.7, 63.3]
Llama-3.1-8B	C2	98.6 [97.7, 99.2]	56.3 [50.7, 61.8]	0.0 [0.0, 0.8]	58.7 [54.4, 63.0]
Llama-3.1-8B	**C3**	**100.0 [99.6, 100.0]**	**88.3 [84.2, 91.5]**	**4.0 [2.6, 6.1]**	**55.9 [51.5, 60.2]**
Llama-3.1-8B	C3-L1	72.4 [69.5, 75.1]	68.0 [62.5, 73.0]	1.0 [0.4, 2.3]	55.7 [51.3, 60.0]
Llama-3.1-8B	C3-L2	100.0 [99.6, 100.0]	82.7 [78.0, 86.5]	5.0 [3.4, 7.3]	54.7 [50.3, 59.0]
Llama-3.1-8B	C3-L3	100.0 [99.6, 100.0]	89.7 [85.7, 92.6]	4.0 [2.6, 6.1]	55.9 [51.5, 60.2]
Llama-3.1-8B	C3-L4	98.7 [97.8, 99.2]	57.7 [52.0, 63.1]	0.0 [0.0, 0.8]	59.5 [55.2, 63.7]
Llama-3.1-8B	C3-ISO	100.0 [99.6, 100.0]	85.7 [81.2, 89.2]	5.8 [4.1, 8.2]	53.5 [49.1, 57.8]